

end, let $\mathcal{D}$ be an agreement tree-child digraph for $\mathcal{N}$ and $\mathcal{N}'$, and let $c_{\mathcal{D}}$ and $c'_{\mathcal{D}}$ be the cut size of $\mathcal{D}$ in $\mathcal{N}$ and $\mathcal{N}'$, respectively. Then $\mathcal{D}$ is called a *maximum agreement tree-child digraph* for $\mathcal{N}$ and $\mathcal{N}'$ if the sum $c_{\mathcal{D}} + c'_{\mathcal{D}}$ is minimum over all agreement tree-child digraphs for $\mathcal{N}$ and $\mathcal{N}'$, in which case we denote this minimum number by $m_{tc}(\mathcal{N}, \mathcal{N}')$. To calculate $m_{tc}(\mathcal{N}, \mathcal{N}')$, it follows from Proposition 4.3 that it is sufficient to consider a single extension of each agreement digraph for $\mathcal{N}$ and $\mathcal{N}'$. Referring back to Figure 1, observe that each of the three phylogenetic digraphs $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$ is in fact an agreement digraph for the two tree-child networks $\mathcal{N}$ and $\mathcal{N}'$ that are shown in the same figure.

Let $\mathcal{N}$ and $\mathcal{N}'$ be two tree-child networks, let $\mathcal{D}$ be an agreement tree-child digraph for $\mathcal{N}$ and $\mathcal{N}'$, and let $\mathcal{R}$ and $\mathcal{R}'$ be an extension of $\mathcal{D}$ in $\mathcal{N}$ and $\mathcal{N}'$, respectively. We note that, similar to the elements of an agreement forest, the elements in $\mathcal{D}$ can be embedded in $\mathcal{N}$ and $\mathcal{N}'$. Intuitively, they can be thought of as subnetworks that are common to $\mathcal{N}$ and $\mathcal{N}'$. On the other hand, the digraphs induced by the edges in $E_{\mathcal{R}} - E_{\mathcal{M}}$ and $E_{\mathcal{R}'} - E_{\mathcal{M}'}$, where $\mathcal{M}$ and $\mathcal{M}'$ is the embedding that underlies $\mathcal{R}$ and $\mathcal{R}'$, respectively, are not necessarily the same. Although each connected component in such a digraph is a rooted tree whose (unique) root is a vertex of $\mathcal{M}$ and $\mathcal{M}'$, respectively, and whose edges are directed *towards* the root, one digraph may contain directed rooted trees with a small total number of unlabelled leaves and the other one may contain directed rooted trees with a much larger total number of unlabelled leaves.

Now, let $\mathcal{N}$ be a phylogenetic network on X , and let $e = (u, v)$ be an edge in $\mathcal{N}$. We consider the following three operations applied to $\mathcal{N}$:

SNPR $^{\pm}$ If u is a tree vertex, then delete e , suppress u , subdivide an edge that is not a descendant of v with a new vertex u' , and add the new edge (u', v) .

SNPR $^{-}$ If u is a tree vertex and v is a reticulation, then delete e , and suppress u and v .

SNPR $^{+}$ Subdivide e with a new vertex v' , subdivide an edge in the resulting network that is not a descendant of v' with a new vertex u' , and add the new edge (u', v') .

By definition of a tree vertex, $u \neq \rho$ if we apply an SNPR $^{\pm}$. If it is not important which of SNPR $^{-}$, SNPR $^{+}$, and SNPR $^{\pm}$ has been applied to $\mathcal{N}$ we simply refer to it as an SNPR. By [6, Proposition 3.1], the operation is *reversible*, i.e. if $\mathcal{N}'$ is a phylogenetic network on X that can be obtained from $\mathcal{N}$ by a single SNPR, then $\mathcal{N}$ can also be obtained from $\mathcal{N}'$ by a single SNPR. Lastly, we note that the well-known rSPR operation is an application of SNPR $^{\pm}$ to a phylogenetic tree.

Let $\mathcal{N}$ and $\mathcal{N}'$ be two phylogenetic networks on X . An SNPR *sequence* σ for $\mathcal{N}$ and $\mathcal{N}'$ is a sequence

$$\sigma = (\mathcal{N} = \mathcal{N}_0, \mathcal{N}_1, \mathcal{N}_2, \dots, \mathcal{N}_t = \mathcal{N}')$$

of phylogenetic networks on X such that, for all $i \in \{1, 2, \dots, t\}$, we have $\mathcal{N}_i$ is obtained from $\mathcal{N}_{i-1}$ by a single SNPR in which case, we say that σ *connects* $\mathcal{N}$ and $\mathcal{N}'$. We refer to t as the *length* of σ . Let $t^{\pm}$ be the number of phylogenetic networks